

MA 765: Topics in Non-linear Algebra

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1 Gröbner Basis

Denote by $\leq$ the coordinate wise partial order on $\mathbb{N}_0^n$. $(a_1, \dots, a_n) \leq (b_1, \dots, b_n)$ if $a_i \leq b_i$ for $i \in [n]$.

Divisibility is a partial order on monomials.

Theorem 1 (Dickson's lemma). Every infinite subset of $\mathbb{N}_0^n$ contains elements a, b with $a < b$.

Proof. The proof follows by induction. Let $\mathcal{M} \subset \mathbb{N}_0^n$ be an infinite subset. For any $i \in \mathbb{N}_0$ define $\mathcal{M}_i$,

$$\mathcal{M}_i = \{a = (a_1, \dots, a_n) \in \mathbb{N}_0^{n-1} : (a, i) \in \mathcal{M}\}$$

If $\mathcal{M}_i$ is finite, then look at $\bigcup_i \in \mathbb{N}_0$, which has finitely many and atleast one minimal element. Thus there is some $j \in \mathbb{N}$ such that $\bigcup_{i=0}^j \mathcal{M}_i$ with $a \in \mathcal{M}_i$ for some $i \leq j$. Hence $(a, i) < (b, k)$. □

Corollary 2. For any $\phi \neq \mathcal{M} \subset \mathbb{N}_0^n$ the set of minimal elements wrt $<$ is nonempty and finite.

3. Monomial order on $\mathbb{N}_0^n$ is a relation $<$ such that

1. If $a \neq b$ then $a < b$ or $b < a$.
2. If $A < b$ and $b < c$ then $a < c$.
3. $(0, \dots, 0) < a$ for any $a \in \mathbb{N}_0^n$.
4. $a < b$ then $a + c < b + c$ for any $c \in \mathbb{N}_0^n$.

First two conditions together imply that $<$ is a total order.

Remark 4. 1. If $a < b$, then $a < b$, i.e., any monomial order refines coordinate wise partial order.

2. Any monomial order on $\mathbb{N}_0^n$ gives total order on monomials in $k[X_n]$

Example 5. Lexicographic order: Left most non zero entry of $a - b$ is positive the $a >_{lex} b$.

degree lex order if either $|a| > |b|$ or $|a| = |b|$ and right most entry of $a - b$ is negative.

degree reverse lex order, same as above but right most entry of $a - b$ is negative.

Proposition 6. For any monomial order $<$ on $\mathbb{N}_0^n$ with $\phi \neq \mathcal{M} \subset \mathbb{N}_0^n$ has a unique minimal element.

Proof. Dickson's lemma gives that $\mathcal{M}$ has a finite non empty set of minimal elements wrt coordinate wise order. The minimal element of these wrt $<$ is the desired element. □

7. Fix a monomial order $<$ on $k[X_n]$

1. The initial monomial $\text{in}_{<}(f)$ of $f = \sum c_a x^a$ is the largest monomial wrt $<$ appearing in f with non zero coefficients.
2. The leading term $\text{lt}_{<}(f) := c_a x^a$ with $x^a = \text{in}_{<}(f)$
3. The initial ideal of an ideal I is $\text{in}_{<}(I) = \langle \text{in}_{<}(f) : f \in I \rangle$.

If G is a generating set for an ideal I then $\langle \text{im}(g) : g \in G \rangle \subset \text{im}(I)$ and this inclusion can be strict.

Proposition 8. Let $<$ be a monomial order on $k[X_n]$. Every ideal I has a finite subset $\mathcal{G}$ such that $\text{im}(I) = \langle \text{im}(g) : g \in \mathcal{G} \rangle$.

Any such $\mathcal{G}$ is called a Gröbner Basis of I wrt $<$.

Proof. The set of monomials in $\text{im}(I)$ has a finite and nonempty subset of minimal elements wrt divisibility, say $m_1, \dots, m_s$. Thus $\text{im}(I) = \langle m_1, \dots, m_s \rangle$. Every monomial in $\text{im}(I)$ is the initial monomial of some $f \in I$. hence there exists $f_1, \dots, f_s \in I$ with $\text{im}(f_i) = m_i$. Thus $\{f_i\}$ is a Gröbner basis. $\square$

Theorem 9. If $\mathcal{G}$ is a Gröbner basis of I , then $I = \langle \mathcal{G} \rangle$

Note that Hilbert's basis theorem is a simple corollary of this.

Proof. We argue by contradiction. By 6, choose $f \in I - \langle \mathcal{G} \rangle$ such that $\text{im}(f)$ is minimal. Call $\text{im}(f) = x^b$. $x^b \in \text{im}(I) = \langle \text{im}(g) : g \in \mathcal{G} \rangle$. There exists $g \in \mathcal{G}$ such that $\text{im}(g) | x^b$, say $x^b = x^c \cdot \text{im}(g)$.

$\text{im}(f - x^c \lambda g) < x^b = \text{im}(f)$ where $\lambda = \text{lt}(f)/x^c \text{lt}(g) \in k$. But $f - x^c \lambda g \in I - \langle \mathcal{G} \rangle$, which is a contradiction to the minimality. $\square$

Lemma 10. Consider $f, g_1, \dots, g_s \in k[X_n]$, with $g_i \neq 0$. Then for any minimal order $<$, there exists $q_1, \dots, q_s, r \in k[X_n]$ such that

1. $f + \sum_{i=1}^s q_i g_i + r$
2. $\text{im}(f) \geq \text{im}(q_i g_i) \forall i$ (note that $\text{im}(f) = \text{im}(q_i g_i)$ for some i)
3. $\text{im}(r)$ is not divisible by $\text{im}(g_i)$ for any i .

We say f reduces to r by $\{g_1, \dots, g_s\}$.

11 Division Algorithm. Input: $f, g_1, \dots, g_s$

Output: $q_1, \dots, q_s, r$ satisfying the properties 1-3.

1. Set $r = 0, p = gf, q_1 = \dots = q_s = 0$
2. While $p \neq 0$ do :

If $\text{im}(g_i)$ divides $\text{im}(p)$ for some $i \in [s]$, then set $q_i = q_i + \frac{\text{lt}(p)}{\text{lt}(g_i)}$ and $p_i = p - \frac{\text{lt}(p)}{\text{lt}(g_i)} g_i$
 else set $r = r + \text{lt}(p), p = p - \text{lt}(p)$
3. Return $q_1, \dots, q_s, r$

Example 12. Consider Lexicographic ordering with $x > y$ and $f = x^2y + xy^2 + y^2$ and

$$g_1 = xy - 1, g_2 = y^2 - 1.$$

p	$xy - 1$	$y^2 - 1$	r
$x^2y + xy^2 + y^2 - x^2y + x$	x		
$xy^2 + y^2 + x - xy^2 + y$	y		
$y^2 + x + y - x$			x
$y^2 + y - y^2 + 1$	1		
$y + 1 - y$			y
1			1

Corollary 13. Buchberger's criteria Let $\mathcal{G}$ be a finite subset of I . Then $\mathcal{G}$ is a Gröbner basis of I iff each $f \in I$ can be reduced to 0 by $\mathcal{G}$.

Proof. If f reduces to r by $\mathcal{G}$, then $\text{im}(g_i)$ does not divide $\text{im}(r)$, for all i . However $f - r \in \langle \mathcal{G} \rangle$ and $r \in I$ and $\mathcal{G}$ is a Gröbner basis of I . So there exists some $g \in \mathcal{G}$ such that $\text{im}(r)$ is divisible by $\text{im}(g)$, which forces $r = 0$.

Conversely, we have $f = \sum q_i g_i$ with $g_i \in \mathcal{G}$ and $\text{im}(q_i g_i) \leq \text{im}(f)$. Hence equality for some i and so $\text{im}(f) \in \langle \text{im}(g) : g \in \mathcal{G} \rangle$. $\square$

Proposition 14. Let $<$ be a monomial order. Then

1. Let B be the set of monomials in $k[X_n] - \text{im}(I)$. Then $\bar{B} \subset k[X_n]/I$ is a k vsp basis.
2. If $\mathcal{G}$ is a Gröbner basis of I , then the remainder of f by $\mathcal{G}$ is unique and does not depend on the choice of $\mathcal{G}$.

Proof. 1. If $p = \sum \lambda_i m_i \in I$ with $m_i \in B$, then $\text{im}(p) \in \text{im}(I)$, but $\text{im}(p) = \text{im}(m_i) \notin I$. To show $\bar{B}$ spans. $m \in k[X_n]$ such that $\bar{m} \notin \text{span}(\bar{B})$.

Take $\min\{m\} = m$ where $m \notin B$. So we have $m \in \text{im}(I)$. There exists $f \in I$ such that $\text{im}(f) = m$. So any monomial in $f - \text{lt}(f) + I = f - \lambda m + I$ is in $\text{span}(\bar{B})$. So $\lambda m + I = p - f + I \in \text{span}(\bar{B})$. This leads to a contradiction $\square$

Definition 15. 1. For terms $\lambda x^a, \mu x^b$ ($\lambda, \mu \in k$) denote by

$$\begin{aligned} \gcd(\lambda x^a, \mu x^b) &= \gcd(x^a, x^b) \\ \text{lcm}(\lambda x^a, \mu x^b) &= \text{lcm}(x^a, x^b) \end{aligned}$$

2. For $0 \neq g, h \in k[x_n]$, their s -polynomial (wrt monomial order $<$)

$$S(g, h) := \frac{\text{lt}(h)}{\gcd(\text{lt}(h), \text{lt}(g))} g = \frac{\text{lt}(g)}{\gcd(\text{lt}(h), \text{lt}(g))} h.$$

16. Buchberger's Algorithm for computing Gröbner basis Input: $f_1, \dots, f_s \in k[X_n]$ in monomial order.

Output: Gröbner basis $\mathcal{G}$ of $I = \langle f_1, \dots, f_s \rangle$ wrt $<$.

1. Set $\eta = \langle f_1 \dots, f_s \rangle$
2. Order the elements of $\mathcal{G}$ as $f_1, \dots, f_t$
3. For $1 \leq i < j \leq t$ do: Reduce $S(g_i, g_j)$ to r by g . If $r \neq 0$, then set $\mathcal{G} := \mathcal{G} \cup \{r\}$ and go to step 2.
4. Return $\mathcal{G}$

Remark 17. The algorithm computes a Gröbner basis. It terminate because in case $r \neq 0$, $\text{im}(r) \notin \langle \text{im}(g_1), \dots, \text{im}(g_t) \rangle$.

18. Extension to submodules of finitely generated $k[X_n]$ module So $F = k[X_n]^r = \bigoplus_{i=1}^r k[X_n]e_i$.

Monomials in G are of the form $x^a e_i$ and terms are $\lambda x^a e_i$ with $\lambda \in k$.

A monomial order on F is a total order on the monomials satisfying:

If $x^a \neq 1$, then $m_1 < m_2 \implies m_1 < x^a \cdot m_1 < x^a m_2$, for monomials m_i in F .

Given a monomial order on the polynomial ring $k[X_n]$ and an order on $\{e_i\}$. We obtain a monomial ordering on F by ordering $\mathbb{N}_0^n \times [r]$ or $[r] \times \mathbb{N}_0^n$ lexicographically.

Dicksen's lemma can be extended to monomials in F . We also define im , lt with respect to $<$ analogously. There is also a division algorithm.

Theorem 19. 1. Every submodule M of F has finite Gröbner basis and the basis generates M .

2. If B is the set of monomials in $F - \text{im}(M)$, then $\bar{B} \subset F/M$ form a k basis of F/M .

2 Hilbert Functions

Definition 20. Let $I \subset k[X_n]$ be a monomial ideal. The Hilbert function of $A = k[X_n]/I$ (or of I) is

$$h_A : \mathbb{N}_0 \rightarrow \mathbb{Z}$$

$$a \mapsto h_A(j) = \dim_k[A]_j$$

where $[A]_j$ is k vector space of images of polynomials of degree j , from $k[X_n] \rightarrow A$. (So $h_A(j)$ = number of monomials in $[k[X_n]]_j - I$).

It's generating function is the Hilbert series

$$H_A(z) = \sum_{j \geq 0} h_A(j) z^j$$

Example 21. 1. For $I = 0$, we get $h_{k[X_n]}(j) = \binom{n+j-1}{j}$ and so

$$H_{k[X_n]}(z) = \sum_{j \geq 0} \binom{n+j-1}{j} z^j = \frac{1}{(1-z)^n}$$

2. If $I = \langle x^a \rangle$, then let $e = \deg(x^a)$.

3.

$$h_{k[X_n]/I}(j) = \begin{cases} h_{k[X_n]}(j) & j < e \\ h_{k[X_n]}(j) - h_{k[X_n]}(j-e) & j \geq e \end{cases}$$

Hence

$$H_{k[X_n]/I} = \frac{1-z^e}{(1-z)^n}$$

Theorem 22. For any proper monomial ideal I of $k[X_n]$, the Hilbert series of $A = k[X_n]/I$ is a rational function of the form

$$H_A(z) = \frac{\kappa_A(z)}{(1-z)^d}$$

with $\kappa_A(z) \in \mathbb{Z}[z]$, $\kappa_A(0) = 1$, $\kappa_A(1) \neq 0$, $d \in \mathbb{N}$. (this expression is unique.)

The dimension of A is

$$\dim A := d$$

and the multiplicity of A (or degree of I) is

$$\deg(I) = \mathcal{H}_A(1) > 0$$

There is a polynomial called Hilbert polynomial p_A of A such that $h_A(j) = p_A(j)$ if $j \gg 0$. If $d = \dim A > 0$, then

$$p_A(z) = \frac{\deg(I)}{(d-1)!} z^{d-1} + \text{lower order terms}$$

$$= h_0(A) \binom{z+d-1}{d-1} + h_1(A) \binom{z+d-2}{d-2} + \dots + h_{d-1}(A) \binom{z}{0}$$

where $h_0(A) = \deg(I)$ and $h_i(A)$ are integers. Note if $p_A(z) \neq 0$ polynomial, then $\deg p_A = d-1 = \dim A - 1$.

Example 23.

$I = 0$, we get $\dim k[X_n] = n$ and the multiplicity of $k[X_n]$ is $1 = \deg I$.

$I = \langle x^a \rangle$ with $\deg x = e$, then $\dim k[X_n]/I = n - 1$ and $\deg(I) = \deg(x^a)$.

Proof of 22. Inclusion exclusion principle: $|\bigcup_{i=1}^s X_i| = \sum_{\phi \neq T \subset [s]} (-1)^{|T|+1} |X_T|$ where $X_t = \bigcap_{i \in T} X_i$

Let $I = \langle m_1, \dots, m_s \rangle = I$, where m_i are monomials. Denote by $X_i(j)$ set of degree j monomials in $\langle m_i \rangle \subset k[X_n]$. hence for $T \subset [s]$, $X_T(j) = \bigcap_{i \in T} X_i(j)$ is the set of deg j monomials that are divisible by $m_T = \text{lcm}(m_i)_{i \in T}$. Define $e_t := \deg m_T$. So

$$|X_T(j)| = \begin{cases} 0 & j < e_t \\ \binom{n-1+j-e_t}{n-1} & j \geq e_t \end{cases}$$

Thus

$$\sum_{j \geq 0} |X_T(j)| z^j = \sum_{j \geq e_t} \binom{n-1+j-e_t}{n-1} z^j = \sum_{k \geq 0} \binom{n-1+k}{n-1} z^{k+e_t} = z^{e_t} \frac{1}{(1-z)^n}$$

Since

$$\begin{aligned} h_A(j) &= \binom{n-1+j}{n-1} - \text{number of deg } j \text{ monomials in } I = \langle m_1, \dots, m_s \rangle \\ &= \binom{n-1+j}{n-1} - \left| \bigcup_{i \in [j]} X_i \right| \end{aligned}$$

We get

$$\begin{aligned} H_A(z) &= \sum_{j \geq 0} h_A(j) z^j = \sum_{j \geq 0} \binom{n-1+j}{n-1} z^j + \sum_{T \in [s]} (-1)^{|T|} \sum_{j \geq 0} |X_T(j)| z^j \\ &= \frac{1}{(1-z)^n} + \sum_{T \in [s]} (-1)^{|T|} \frac{z^{e_T}}{(1-z)^n} =: \frac{g(z)}{(1-z)^n} \end{aligned}$$

When $g(z) \in \mathbb{Z}[z]$, with $g(0) = 1$ writing $g(z) = (1-z)^v \kappa_A(z)$ with $\kappa_A(z) \in \mathbb{Z}[z]$, suitable $v \in \mathbb{N}$ and $\kappa_A(1) \neq 0, \kappa_A(0) = 1$. Hence $H_A(z) = \frac{\kappa_A(z)}{(1-z)^d}$ where $d = n - v$. Write

$$\kappa_A(z) = \sum_{k=0}^w c_k z^k \text{ with } c_k \in \mathbb{Z}$$

$$\sum_{j \geq 0} h_A(j) z^j = \left(\sum_{k=0}^w c_k z^k \right) \left(\sum_{l \geq 0} \binom{d-1+l}{d-1} z^l \right)$$

comparing coefficients in $\deg j \gg 0$, we get

$$\begin{aligned}
h_A(j) &= \sum_{k+l=j} c_k \binom{d-1+l}{d-1} = \sum_{k=0}^w c_k \underbrace{\binom{d-1+j-k}{d-1}}_{\text{polynomial in } j-k \text{ variables of deg } d-1} \\
&= \underbrace{\left(\sum_{k=0}^w c_k \right)}_{\kappa_A(1)} \binom{j}{d-1} + \text{lower order terms} \\
&=: p_A(j) \text{ (Hilbert polynomial)}
\end{aligned}$$

If $h_A(j) = 0$ whenever $j \gg 0$, then by definition $0 = d = \dim A$ and in this case p_A is the zero polynomial. Hence if $d > 0$, then $h_A(j) > 0$ if $j \gg 0$ and so the leading coefficient of $p_A(z)$ must be positive, i.e., $\kappa_A(1) > 0$ $\square$

Definition 24. A monomial order is called degree compatible if

$$\deg(x^a) > \deg(x^b) \implies x^a > x^b$$

for any two monomials.

25. For any ideal $I \subset k[X_n]$ and any $t \in \mathbb{Z}$, set

$$I_{\leq t} = \{f \in I : \deg f \leq t\}$$

It is a k -subspace of $k[X_n]$. Write $\text{Mon}(k[X_n])$ for the set of monomials in $k[X_n]$.

Lemma 26. Let $<$ be a degree compatible monomial order. For any ideal $I \subset k[X_n] = S$, one has

$$\begin{aligned}
\dim_k \frac{k[X_n]_{\leq t}}{I_{\leq t}} &= \text{number of monomials in } \frac{k[X_n]_{\leq t}}{\text{im}_{<}(I)} \\
&= |\text{Mon}(k[X_n])_{\leq t} - \text{im}_{<}(t)|
\end{aligned}$$

Proof. $B := \text{Mon}(k[X_n])_{\leq t} - \text{im}_{<}(I)$. We claim

$$\bar{B} \subset \frac{k[X_n]_{\leq t}}{I_{\leq t}} \text{ is a } k\text{-basis}$$

$\overline{\text{Mon}(k[X_n]) - \text{im}(I)}$ is a k -basis of $k[X_n] - \text{im}(I)$, by 14. $\bar{B}$ spans remainder of any $F \in k[X_n]$ upon dividing by Gröbner basis of I wrt $<$ satisfies $\deg r \leq \deg f$. $\square$

For $I \subset k[X_n]$, the affine Hilbert function of $A = k[X_n]/I$ is

$$\begin{aligned}
h_A^a : \mathbb{N}_0 &\longrightarrow \mathbb{Z} \\
j &\mapsto h_A^a(j) = \dim_k \frac{k[X_n]_{\leq j}}{I_{\leq j}}
\end{aligned}$$

Lemma 27. For any degree compatible monomial order $<$ on I , one has

$$h_{\frac{k[X_n]}{\text{im}_{<}(I)}}(j) = h_A^a(j) - h_A^1(j-1)$$

where $A = k[X_n]/I$.

Proof. By definition, we have

$$h_{\frac{k[X_n]}{\text{im}(I)}}(j) = |\text{Mon}(k[X_n])_j - \text{im}(I)|$$

$$26 \implies h_A^a(j) = |\text{Mon}(k[X_n])_{\leq j} - \text{im}(I)| \quad \square$$

Remark 28. If $A \neq 0$, $h_A^a(0) = 1$. Then by 27 we have $h_A^a(j) = \sum_{k=0}^j h_{k[X_n]/\text{im}(I)}(k)$. It follows that generating function of h_A^a and $h_{\frac{k[X_n]}{\text{im}(I)}}$ have analogous properties

29. We define dimension of A ,

$$\dim \frac{k[X_n]}{I} := \dim_k \frac{k[X_n]}{\text{im}(I)}$$

and the degree

$$\deg(I) = \deg(\text{im}(I))$$

where $<$ is a degree compatible monomial order.

30. Let $G = (G, +)$ be an abelian group. A G -graded ring R is a family of subgroups $([R]_{a \in G}) \leq (R, +)$ such that

1. $R = \bigoplus_{a \in G} [R]_a$ (as $\mathbb{Z}$ -modules)
2. $[R]_a \cdot [R]_b \subset [R]_{a+b}$

The elements of $[R]_a$ are called homogeneous of degree a .

Example 31. Fine or $\mathbb{Z}$ -grading of $k[X_n]$, where $G = \mathbb{Z}^n$

$$[S]_a = \begin{cases} 0 & \text{if some } a_i < 0 \\ \{\lambda x^a : \lambda \in k\} & \end{cases}$$

Definition 32. $R = G$ -graded ring

1. G -graded R -module is an R -module M with a decomposition $([M]_a)_{a \in G}$ such that $M = \bigoplus_{a \in G} [M]_a$ and $[R]_a \cdot [M]_b \subset [M]_{a+b}$.
2. A G -graded or simply graded submodule of such a graded M is a graded submodule $N \subset M$ such that

$$[N]_a \subset [M]_a$$

Lemma 33. For an arbitrary submodule N of a G graded R -module M the following are equivalent

1. N is a graded submodule
2. N has a generating set consisting of homogeneous elements
3. If $m = \sum_{a \in G} m_a$ with $m_a \in [M]_a$, then $m \in N$ iff each $m_a \in N$.
4. M/N is a G -graded R -module with grading $[M/N]_a := \frac{[M]_a + N}{N}$.

Proof. □

Example 34. 1. M and N are G -graded, then so is $M \oplus N$ with grading $[M \oplus N]_a := [M]_a \oplus [N]_a$ (as R modules). So if R is G graded, then so is R^n .

2. $I \subset k[X_n]$ is a $\mathbb{Z}^n$ graded submodule if I is a monomial ideal.

3. A $\mathbb{Z}$ -graded or homogeneous ideal of $k[X_n]$, is an ideal that has generating set consisting of homogeneous polynomials. In that case $k[X_n]/I$ is a graded module.

35. A homomorphism of G -graded modules or a G -graded homomorphism is a R -module homomorphism $\phi : M \rightarrow N$ that is degree preserving, $\phi([M]_a) \subset [N]_a$.

For any $a \in G$ and a G -graded module M , the module $M(a)$ has the same module structure as M , but grading given by

$$[M(a)]_b := [M]_{a+b}$$

$M(a)$ is a degree a shift of M . (Note here that the convention is opposite of that in algebraic topology.)

Example 36. 1. Consider $k[X_n]$ with standard grading.

$$\begin{aligned} \phi : k[X_n] &\rightarrow k[X_n] \\ f &\mapsto x_1^2 f \end{aligned}$$

is not a graded homomorphism. However define

$$\begin{aligned} \psi : k[X_n](-2) &\rightarrow k[X_n] \\ f &\mapsto x_1^2 f \end{aligned}$$

then $f \in k[X_n](-2)$ has degree $\deg f + 2$. So $f \in [k[X_n](-2)]_{\deg f + 2}$.

2. For any $a \neq 0$ in G , $R(a)$ is not a graded ring, (because identity is not in 0 dimension), but it is a graded R -module.

Lemma 37. If $\phi : M \rightarrow N$ is a homomorphism of graded modules, then $\ker \phi$, $\text{im} \phi$, $\text{coker} \phi$ are graded modules.

Proof. $[\ker \phi]_a = \ker \phi \cap [M]_a$, and $[\text{im} \phi]_a = \text{im} \phi \cap [N]_a$.

□

Example 38. 1. If M is a $\mathbb{Z}$ graded module with generators $m_1, \dots, m_t$ where $d_i = \deg m_i$, then

$$\begin{aligned} \phi : \bigoplus_{i=1}^t R(-d_i) &\longrightarrow M \\ \begin{bmatrix} r_1 \\ \vdots \\ r_t \end{bmatrix} &\mapsto \sum_{i=1}^t r_i m_i \end{aligned}$$

is a homomorphism of graded R -modules and is surjective.

2. Consider $I = \langle x^3, xy, y^4 \rangle \subset k[x, y] + S$ with standard grading.

$$\phi : S(-3) \oplus S(-2) \oplus S(-4) \rightarrow I$$

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \mapsto f_1 x^3 + f_2 xy + f_3 y^4$$

$$\ker \phi : \left\langle \begin{bmatrix} y \\ -x^2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ y^3 \\ -x \end{bmatrix} \right\rangle \xleftarrow[\text{graded hom}]{\cong} S(-4) \oplus S(-5)$$

There exists an exact sequence,

$$0 \rightarrow \begin{array}{c} S(-4) \\ \oplus \\ S(-5) \end{array} \xrightarrow{\begin{bmatrix} y & 0 \\ -x^2 & y^3 \\ 0 & -x \end{bmatrix}} \begin{array}{c} S(-3) \\ \oplus \\ S(-2) \\ \oplus \\ S(-4) \end{array} \xrightarrow[\begin{bmatrix} x^3 & xy & y^4 \end{bmatrix}]{\phi} S \rightarrow S/I \rightarrow 0$$

Definition 39. For any $\mathbb{Z}$ graded module M over $k[X_n]$ its Hilbert function is

$$h_m : \mathbb{Z} \rightarrow \mathbb{Z}$$

$$j \mapsto h_m(j) := \dim_k [M]_j$$

assuming $[M]_j$ is finitely generated for all j .

Remark 40. For any monomial ideal $I \subset k[X_n]$, 20 and 39 agree.

$$\dim_k [k[X_n]/I]_j = |\text{Mon}(k[X_n])_j - I|$$

Proposition 41. For every graded submodule M of a finitely generated free $k[X_n]$ module F and any monomial order $<$ of F ,

$$h_{F/M}(j) = \frac{h_F(j)}{\text{im}(M)} \quad \forall j \in \mathbb{Z}$$

Corollary 42. For any finitely generated $k[X_n]$ submodule $m \neq 0$, Hilbert series is of the form

$$H_M(z) := \sum_{j \in \mathbb{Z}} h_M(j) z^j$$

$$= \frac{\kappa_M(z) z^t}{(1-z)^d}$$

where $\kappa_M(z) \in \mathbb{Z}[z]$, $\kappa_M(0) \neq 0$, $\kappa_M(1) > 0$, $t \in \mathbb{Z}$.

There is a Hilbert polynomial $p_M(z) \in \mathbb{Q}(z)$ such that

$$h_M(j) = p_M(j) \quad j \gg 0$$

(krull) dimension of M is defined as

$$\dim M = d$$

and the degree is

$$\deg(M) = \kappa_M(1)$$

Proof. Let M be generated by $m_1, \dots, m_t$ of degree $d_1, \dots, d_t$ respectively. Define

$$\begin{aligned} \phi : \bigoplus_{i=1}^t S(-d_i) &\twoheadrightarrow M \\ f &\mapsto x_1^2 f \end{aligned}$$

is not a graded homomorphism. However define

$$\begin{aligned} \psi : k[X_n](-2) &\rightarrow k[X_n] \\ \begin{bmatrix} f_1 \\ \vdots \\ f_t \end{bmatrix} &\mapsto \sum f_i m_i \end{aligned}$$

Set $N = \ker \phi$ and we have a short exact sequence $0 \rightarrow N \rightarrow F \rightarrow M \rightarrow 0$ and by rank nullity theorem we have $h_M(j) = h_F(j) - h_N(j)$. S and F have desired Hilbert series which are rational functions. So it is enough to show the same for N , which is same as $h_{F/\text{im}(N)}$.

$\text{im}(N)$ is generated by monomials. So

$$\frac{F}{\text{im}(N)} \cong \bigoplus_{i=1}^t \underbrace{\left(\frac{S}{J_i} \right)}_{\text{has the desired properties}} (-d_i)$$

for monomial ideals J_i . □

Example 43. 1. $S = k[X_n], F = \bigoplus_{i=1}^t S(-d_i)$. Then

$$H_{S(-d_i)}(z) = \frac{z_i^d}{(1-z)^n}$$

2. Consider $I = \langle x^3, xy, y^4 \rangle \subset k[x, y] + S$ with standard grading. $A = S/I$. We have the exact sequence,

$$\begin{array}{ccccccc} & & & S(-3) & & & \\ & & & \oplus & & & \\ 0 \rightarrow & S(-4) & \rightarrow & S(-2) & \xrightarrow{\phi} & S & \rightarrow A = S/I \rightarrow 0 \\ & \oplus & & \oplus & & \begin{bmatrix} x^3 & xy & y^4 \end{bmatrix} & \\ & S(-5) & & S(-4) & & & \end{array}$$

$$\begin{aligned} H_A(z) &= H_S(z) - H_{S(-3) \oplus S(-2) \oplus S(-4)}(z) + H_{S(-4) \oplus S(-5)}(z) \\ &= H_S(z) - H_{S(-3)}(z) - H_{S(-2)}(z) + H_{S(-4)}(z) + H_{S(-5)}(z) - H_{S(-4)}(z) \\ &= \frac{1 - z^3 - z^2 + z^5}{(1-z)^2} \\ &= 1 + 3z + 2z^2 + z^3 \end{aligned}$$

So $\dim A = 0$ and $\deg M = 6$ (polynomial evaluated at 1).

Lemma 44. Let $d \in \mathbb{N}$. Every $a \in \mathbb{N}$ admits a unique presentation of the form

$$a = \binom{k_d}{d} + \binom{k_{d-1}}{d-1} + \cdots + \binom{k_s}{s}$$

with integers $k_d > k_{d-1} \cdots > k_s$. It is called the d -Macaulay presentation of a .

Example 45. For $d = 3$ and $a = 12$, we have

$$12 = \binom{5}{3} + \binom{2}{2} + \binom{1}{1}$$

Definition 46. If $a > 0$ with d -Macaulay presentation

$$a = \binom{k_d}{d} + \binom{k_{d-1}}{d-1} + \cdots + \binom{k_s}{s}$$

set

$$a^{(d)} = \binom{k_{d+1}}{d+1} + \binom{k_d}{d} + \cdots + \binom{k_{s+1}}{s+1}$$

Example 47. $12^{(3)} = 17$

Theorem 48 (Macaulay). Let $h : \mathbb{N}_0 \rightarrow \mathbb{Z}$, the following are equivalent,

1. There is some $n \in \mathbb{N}_0$ and some homogeneous ideal $I \subset k[X_n]$ such that Hilbert function of $A = k[X_n]/I$ is h .
2. There is a monomial ideal (Lexicographic ideal) $I \subset k[X_n]$ with $n = h(1)$ such that Hilbert function of A is h .
3. $h(0) = 1$ and

$$h(j+1) \leq h(j)^{(j)} \text{ if } j > 0$$

Moreover for every graded k -algebra A , one has

$$h_A(j+1) = h_A(j)^{(j)} \text{ if } j \gg 0$$

Example 49.

j	0	1	2	3	4	5	6
$\tilde{h}$	1	4	10	12	18	18	...
h	1	4	10	12	17	17	...

$\tilde{h}$ is not a possible Hilbert function because $12^{(3)} = 17$. While h is a possible Hilbert function.

3 Ideals and Schemes

3.1 Affine case

Definition 50. A ring R is reduced if $r^n = 0$ for some $n \in \mathbb{N}$ implies $r = 0$.

Lemma 51. Consider a reduced ring R and an ideal I of R . Then R/I is reduced iff $I = \sqrt{I}$.

Hilbert's Nullstellensatz gives bijection if $k = \bar{k}$. Let $S = k[x_1, \dots, x_n]$.

$$\left\{ \begin{array}{c} \text{Subvarieties of} \\ \mathbb{A}_k^n \end{array} \right\} \xleftrightarrow[\mathbb{Z}]{I} \left\{ \begin{array}{c} \text{Radical Ideals of} \\ S \end{array} \right\} \xleftrightarrow{\quad} \left\{ \begin{array}{c} \text{Reduced factor} \\ \text{rings of } S \end{array} \right\}$$

$$J \mapsto S/J$$

$$\ker(S \rightarrow A) \leftarrow A$$

Definition 52. The geometric object X associated to an ideal $J \subset S$ is called an affine (sub) scheme of $\mathbb{A}_k^n$. $I_X := J$ is called the defining ideal of X and S/J is called co-ordinate ring. $X = \text{spec}(S/J)$ to denote the scheme X . The reduced subscheme of X is $X_{\text{red}} = \text{spec}(S/\sqrt{J})$. It is also called the *support* of X .

Remark 53. 1. Definition 52 is a special case of an affine scheme. $\text{Spec}(S/J)$ is the set of prime ideals of S/J endowed with Zariski topology where closed sets are of the form $V(p)$ where p is a prime in S containing J .

2. If $k = \bar{k}$, then the points of $X = \text{Spec}(k[X_n]/\sqrt{J}) \subset \mathbb{A}_k^n$ are the points of $X_{\text{red}} = Z(J) = Z(\sqrt{J})$. (The scheme X captures more information, for example multiplicities of the common zeroes)

Example 54. For any $j \in \mathbb{N}$ the scheme $Y_j \subset \mathbb{A}^n$ defined by $(x_1, \dots, x_n)^j$ is supported at the point $(0, \dots, 0)$ i.e. $(Y_j)_{\text{red}} = \{(0, \dots, 0)\}$. Sometimes Y_j is called a *fat point*.

Definition 55. The dimension of $Y = \text{Spec}(k[X_n]/J)$ is $\dim Y = \dim k[X_n]/J$ (as defined using Hilbert series 22)

Example 56. $\dim \text{Spec} \frac{k[X_n]}{(x_1, \dots, x_n)^j} = 0 \quad \forall j$

Definition 57. Let $X, Y \subset \mathbb{A}^n$ be subschemes of $\mathbb{A}^n$. Then X is called a subscheme of Y , if $I_Y \subset I_X$. In symbols $X \subset Y$.

The intersection $X \cap Y$ is the scheme defined by $I_X + I_Y$ and the union $X \cup Y$ is defined by $I_X \cap I_Y$.

Example 58. Continuing with the notation used in 54, we have $Y_1 \subsetneq Y_2 \subsetneq \dots$, but $Y_2 \not\subset \text{Spec}(\frac{k[X_n]}{(x_2, \dots, x_n)})$ ("fat point sticks out of line")

Theorem 59 (Primary decomposition theorem).

Example 60. An affine scheme $Y \subset \mathbb{A}^n$ is irreducible if for any subschemes $Y_1, Y_2 \subset Y$ with $Y_1 \cup Y_2 = Y$ either $Y = Y_1$ or $Y = Y_2$ or $Y_{\text{red}} = (Y_1)_{\text{red}} = (Y_2)_{\text{red}}$

Lemma 61. 1. Y is irreducible iff I_Y is primary.

2. Y is irreducible and reduced iff I_Y is a primary ideal.

62. 1. The fat points Y_j in 54 are irreducible but not reduced if $j \geq 2$

2. A line Y is defined by $I_Y = \langle l_1, \dots, l_{n-1} \rangle$, where l_i are linear independent polynomials in $k[X_n]$. Any line is reduced and irreducible.

Corollary 63. Every affine scheme is a finite union of irreducible schemes

3.2 Projective Schemes

64 Notation. $m = \langle x_0, \dots, x_n \rangle \subset k[x_0, \dots, x_n] \subset S$.

$J \subsetneq S$ is a homogeneous ideal or equivalently $J \subset m$

If $k = \bar{k}$ we have bijections

$$\left\{ \begin{array}{c} V \subset \mathbb{P}^n \\ \text{projective} \\ \text{variety} \end{array} \right\} \xrightleftharpoons[Z]{I} \left\{ \begin{array}{c} \text{homogeneous} \\ \text{Radical Ideals} \\ J \subset m \end{array} \right\} \cong \left\{ \begin{array}{c} \text{Reduced graded} \\ \text{quotient} \\ \text{rings of } S \end{array} \right\}$$

Definition 65. Let $\mathfrak{p}$ be a prime ideal. A $\mathfrak{p}$ -primary ideal $\mathfrak{q}$ is a primary ideal $\mathfrak{q}$ with $\sqrt{\mathfrak{q}} = \mathfrak{p}$

Lemma 66. A homogeneous ideal $J \subset m$ is m -primary iff $\sqrt{J} = m$.

Definition 67. The saturation of a homogeneous ideal J is the ideal

$$J^{\text{sat}} := \bigcup_{n \geq 1} (J : m^n) \supset J$$

J is saturated if $J = J^{\text{sat}}$.

Lemma 68. Let $J \subset m$ be homogeneous. TFAE

1. J is saturated
2. m is not an associated prime ideal of S/J .
3. There is some homogeneous $f \in S$ of positive degree such that $\bar{f} \in S/J$ is a non-zero divisor equivalent to $(J : f) = J$.

Remark 69. 1. If $J = q_1 \cap q_2 \cap \dots \cap q_s$ is a minimal primary decomposition of J with homogeneous q_i and say q_s is m -primary, then

$$J^{\text{sat}} = q_1 \cap \dots \cap q_{s-1}$$

2. If $\sqrt{J} \subsetneq m$ then J^{sat} is the largest homogeneous ideal $I \subset S$ such that $[J]_k = [I]_k$ for any $k \gg 0$. ($[J] - k$ is the space of polynomials of degree k .)

Example 70. Consider $J = \langle x^3, x^2y \rangle = \langle x^2 \rangle \cap \underbrace{\langle x^3, y \rangle}_{\langle x, y \rangle}$. $J^{\text{sat}} = \langle x^2 \rangle$. In $k[x, y, z]$, $J^{\text{sat}} = J$

Definition 71. For every homogeneous ideal J with $\sqrt{J} \subsetneq m$, we consider S/J^{sat} as a geometric object X called a projective (sub)scheme of $\mathbb{P}^n$. The homogeneous ideal of X is $I_X = J^{\text{sat}}$ and S/J^{sat} is called the homogeneous coordinate ring of X . Sometimes we write $X = \text{Proj}(S/J) = \text{Proj}(S/J^{\text{sat}})$ for the projective scheme defined by J .

Remark 72. 1. One has the following bijections

$$\{\emptyset\} \cup \left\{ \begin{array}{c} \text{Projective} \\ \text{subschemes} \\ \text{of } \mathbb{P}^n \end{array} \right\} \xleftrightarrow{1:1} \{m\} \cup \left\{ \begin{array}{c} \text{homogeneous} \\ \text{saturated ideals} \\ J \\ \text{with } \sqrt{J} \subsetneq m \end{array} \right\} \xleftrightarrow{1:1} \{ \cong S/m \} \cup \left\{ \begin{array}{c} \text{graded quotient} \\ \text{ring} \\ \text{of } S \text{ with a} \\ \text{non-zero} \\ \text{divisor of} \\ \text{positive degree} \end{array} \right\}$$

2. For projective subschemes $X, Y \subset \mathbb{P}^n$ the concepts of reducible, irreducible, $X \subset Y$, $X \cap Y$, $X \cup Y$ are analogous to affine case.

$$\begin{aligned} I_{X \cap Y} &= (I_X + I_Y)^{\text{sat}} \\ I_{X \cup Y} &= (I_X \cap I_Y) \end{aligned}$$

Example 73. $X, Y \subset \mathbb{P}^3$ be schemes with homogeneous ideals

$$I_X = \langle x_0, x_1 \rangle \cap \langle x_2, x_3 \rangle \sim \text{pair of skew lines}$$

$$I_Y = \langle x_1 + x_2 \rangle \sim \text{hyperplane}$$

$X \cap Y$ should consist of two points.

$$I_X + I_Y = \langle x_0, x_1, x_2 \rangle \cap \langle x_1, x_2, x_3 \rangle \cap \underbrace{\langle x_0, x_3, x_1 + x_2, x_1 x_2 \rangle}_{\langle x_0, \dots, x_3 \rangle - \text{primary}}$$

$$I_{X \cap Y} = (I_X + I_Y)^{\text{sat}} = \langle x_0, x_1, x_2 \rangle \cap \langle x_1, x_2, x_3 \rangle \text{ and } X \cap Y = X \cap Y_{\text{red}} = \{(0 : 0 : 0 : 1), (1 : 0 : 0 : 0)\}.$$

$$\dim X = 1, \dim Y = 2$$

$$\deg X = 2, \deg Y = 1$$

Definition 74. For a projective subscheme $X \subset \mathbb{P}^n$, we define it's dimension as

$$\dim(X) = \dim(S/I_X) - 1$$

and degree as

$$\deg X = \deg I_X = \deg(S/I_X)$$

4 Bezout's theorem

Definition 75. Consider any R -module M . An element $r \in R$ is called M -regular if $rm = 0$ for any $m \in M \implies m = 0_M$. Otherwise r is called a zero-divisor of M .

Note. $f \in R$ is M -regular if $0 :_M f = 0 :_M \langle f \rangle = 0$

Example 76. The zero divisors of $\mathbb{Z} \text{-- mod } \mathbb{Z}/6\mathbb{Z}$ are precisely the integers $\langle 2 \rangle \cup \langle 3 \rangle$

Proposition 77. If M is finitely generated graded S -module and $f \in S$ is M regular of positive degree, then

1. $\dim M/fM = \dim M - 1$
2. $\deg M/fM = \deg f \deg M$

Proof. Since f is M regular, there exists a short exact sequence ($d = \deg f$)

$$\begin{aligned} 0 \rightarrow M(-d) \xrightarrow{f} M \rightarrow \frac{M}{fM} \rightarrow 0 \\ m \mapsto fm \end{aligned}$$

So $h_{M/fM}(j) = h_M(j) - h_M(j-d)$ □

Lemma 78 (Mayer-Vietoris sequence). If N_1, N_2 are graded submodules of a graded submodule M , then there is a SEQ of graded modules

$$\begin{aligned} 0 \rightarrow \frac{M}{N_1 \cap N_2} \rightarrow \frac{M}{N_1} \oplus \frac{M}{N_2} \rightarrow \frac{M}{N_1 + N_2} \rightarrow 0 \\ m \mapsto (m, m) \\ (m_1, m_2) \mapsto m_1 - m_2 \end{aligned}$$

Corollary 79. If $I, J \subset S$ are homogeneous ideals, then $h_{\frac{S}{I+J}}(k) = h_{\frac{S}{I}}(k) + h_{\frac{S}{I \cap J}}(k)$ for all $k \in \mathbb{Z}$.

Corollary 80. 1. $\dim(\frac{S}{I \cap J}) = \max\{\dim S/I, \dim S/J\}$

2. If WLOG $\dim S/I \geq \dim S/J$, then

$$\deg\left(\frac{S}{I \cap J}\right) = \begin{cases} \deg S/I & \dim S/I > \dim S/J \\ \deg S/I + \deg S/J & \dim S/I = \dim S/J > \dim S/I + J \\ \deg S/I + \deg S/J & \dim S/I = \dim S/I + J \\ -\deg S/I + J & \end{cases}$$